GHOST-PATH

AD ENUMERATION TOOL

PREPARED FOR:

ENG. AHMED ESSAM

PREPARED BY:

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1. Project Planning & Management

1.1 Project Proposal

- **Project Overview**: GhostPath is an advanced Active Directory (AD) enumeration tool designed for security professionals and red teamers. It serves as a specialized scanner to gather critical information from a domain environment through a professional, high-contrast command-line interface.
- **Objectives**:
  - To provide an automated and structured method for enumerating AD objects (Users, Machines, and Groups).
  - To enhance the visibility of security risks by identifying Kerberoastable accounts and privileged domain groups.
  - To deliver a user-friendly experience with organized, color-coded output for rapid data analysis.
- **Project Scope**: The project involves refactoring the original enumeration logic, optimizing LDAP queries for speed, and designing a high-fidelity CLI reporting system. It covers user property inspection, machine operating system categorization, and group membership mapping.

1.2 Project Plan (Milestones & Deliverables)

The project was executed following a structured timeline to ensure all technical requirements were met:

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4. **Phase 4: Documentation**: Completing the technical manuals and GitHub repository structure.

1.3 Task Assignment & Roles

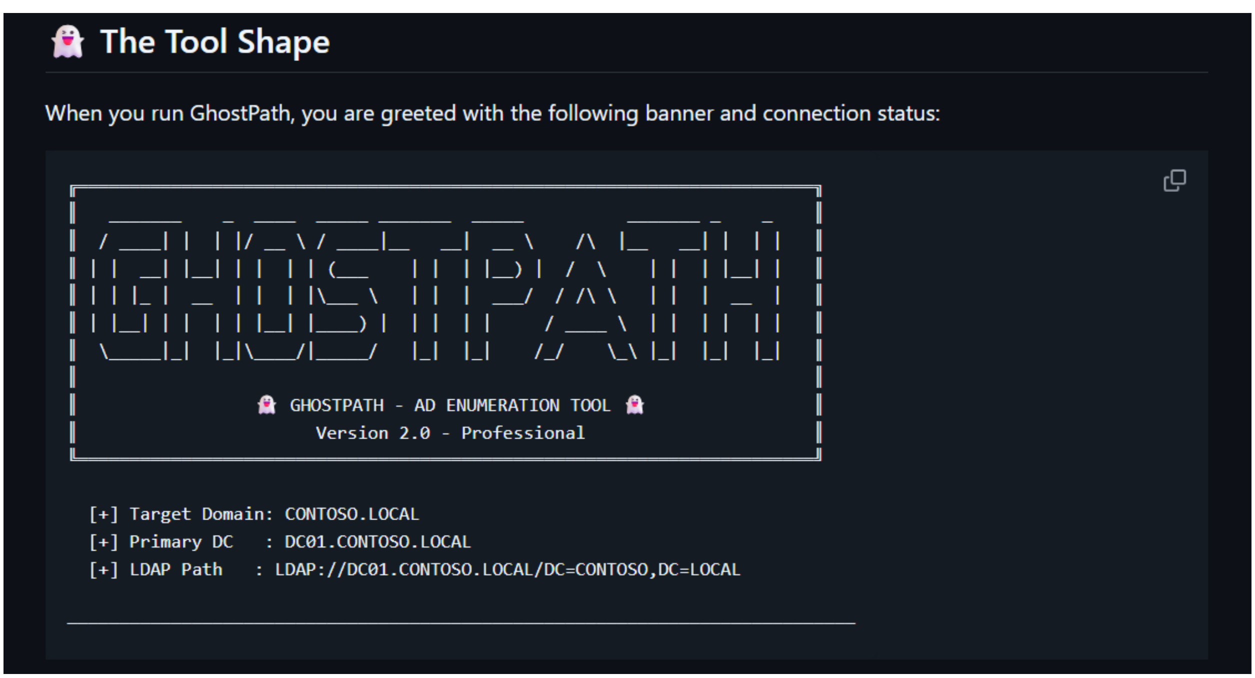
- **Security Developers**: Responsible for full-stack development, logic refactoring, and UI design.
- **Security Analyst & Tester**: Responsible for defining security test cases, verifying LDAP outputs, and performing risk assessments.

1.4 Risk Assessment & Mitigation Plan

Identified Risk	Mitigation Strategy
Detection by AV/EDR: Security tools might flag the enumeration activity.	Use standard LDAP queries and native Windows protocols to blend with legitimate traffic.
Performance Lag: Large AD environments may cause slow data retrieval.	Optimized LDAP filters to fetch only required attributes, reducing network overhead.
Permission Issues: Tool might fail if run with low-privileged users.	Implemented clear error handling to notify the user of insufficient permissions.

1.5 Key Performance Indicators (KPIs)

- **Execution Speed**: The tool should complete a full domain scan (Users, Machines, Groups) in under 60 seconds for mid-sized environments.
- **Data Accuracy**: 100% accuracy in identifying "Domain Admins" and "Kerberoastable" accounts.
- **Usability**: The high-contrast interface should allow a user to identify the Domain Controller and target users within 5 seconds of looking at the output.
- **Stability**: Zero crashes during the enumeration of objects with null or unusual attributes.



2. Requirements Gathering

2.1 Stakeholder Analysis

The development of GhostPath involved identifying the key users and entities that will interact with or benefit from the tool:

- **Primary Users**: Red Teamers, Penetration Testers, and Security Auditors who require a fast and readable way to map Active Directory environments.
- **Secondary Users**: System Administrators who want to audit domain privileges and identify misconfigured or vulnerable service accounts.
- **Infrastructure**: The Active Directory Domain Controller (DC) which serves as the data source via LDAP.

2.2 User Stories & Use Cases

To ensure the tool meets real-world needs, the following user stories were defined:

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These requirements define the specific behaviors and functions of the GhostPath tool:

1. **FR1: User Enumeration**: The tool must be able to list all domain users and extract key security attributes (SPNs, AS-REP roasting status).
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5. **FR5: Property Inspection**: The tool must support a "Detailed View" mode to display all LDAP properties of a specific target object.

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These requirements define the performance and quality standards of the tool:

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2. **NFR2: Usability (UI/UX)**: The command-line interface (CLI) must use high-contrast text and ASCII-styled tables to ensure maximum readability for the operator.
3. **NFR3: Portability**: The tool must run as a standalone script or executable on any Windows-based system with access to the domain, without requiring complex installations.
4. **NFR4: Reliability**: The tool must include robust error handling to manage scenarios where the Domain Controller is unreachable or permissions are restricted.

2.5 System Constraints

- **Environment**: Must be executed from a domain-joined machine or with valid domain credentials.
- **Network**: Requires an active network path to the Domain Controller via LDAP (Port 389) or LDAPS (Port 636).
- **Execution Policy**: Requires appropriate PowerShell execution permissions or a compiled executable.

3. System Analysis & Design

3.1 Problem Statement & Objectives

- **Problem Statement**: Manual Active Directory enumeration is time-consuming and often results in unorganized data. Security professionals need a tool that can parse LDAP data into a readable and actionable format without the overhead of complex database setups.
- **Project Goals**:
  - Automate the retrieval of AD objects (Users, Machines, Groups).
  - Provide a high-contrast visual interface for better data clarity.
  - Identify high-risk accounts (Kerberoastable) and privileged groups automatically.

3.2 Software Architecture

The system follows a **Modular Standalone Architecture** (Monolithic), consisting of three primary layers:

1. **Input Layer (CLI)**: Responsible for receiving user commands and parameters (-ObjType, -Name).
2. **Logic & Processing Layer**: The core engine that translates user requests into specific LDAP filters and processes the raw data received from the Domain Controller.
3. **Presentation Layer**: A dedicated UI module that formats the processed data into ASCII tables and high-contrast banners for the terminal.

3.3 Database Design & Data Modeling

- **Note on Database**: GhostPath is a stateless tool. It does **not** use a local database (like SQL or Neo4j).
- **Data Source**: It interacts directly with the **Active Directory Database (NTDS.dit)** via the **LDAP protocol**.
- **Data Modeling**: Data is treated as transient objects during the session. Objects are mapped from LDAP attributes (sAMAccountName, servicePrincipalName) into structured custom objects within the tool's memory for display.

3.4 System Behavior

- **Use Case Diagram**:
  - **Actor**: Security Professional / Auditor.
  - **Actions**: Run Enumeration, Filter by Object Type, Inspect Specific Properties, Identify Vulnerable Accounts.
- **Data Flow Diagram (DFD - Level 0)**:
  - **Source**: User Input -> **Process**: GhostPath Tool -> **Destination**: Active Directory via LDAP.
  - **Return Path**: Active Directory -> GhostPath Tool (Filtering/Formatting) -> User (Terminal Display).
- **Sequence Diagram**:
  1. User executes GhostPath.ps1 with parameters.
  2. Tool initiates a connection to the Primary Domain Controller (PDC).
  3. PDC authenticates the request and returns requested LDAP objects.
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3.5 UI/UX Design & Prototyping

- **Design Principles**:
  - **Readability**: Use of structured tables with clear headers.
  - **Emphasis**: Using visual cues like [!] for privileged groups and [B] for built-in groups.
  - **High Contrast**: A professional ASCII banner to distinguish different sections of the scan results.
- **Color Palette**: The UI is designed for dark-mode terminals, ensuring that highlighted alerts are easily visible.

3.6 Technology Stack

- **Development Environment**: Visual Studio / VS Code.
- **Languages**: PowerShell (for Active Directory ADSI/LDAP integration).
- **Protocols**: LDAP (Lightweight Directory Access Protocol) on ports 389/636.

4. Implementation (Source Code & Execution)

4.1 Source Code Quality & Standards

The implementation of GhostPath adhered to high-quality software development standards to ensure maintainability and performance:

- **Modular Programming**: The source code is organized into logical modules, separating the LDAP query engine from the UI rendering logic.
- **Coding Standards**: Followed consistent naming conventions (PascalCase for functions and camelCase for variables) to ensure code readability.
- **Error Handling**: Implemented robust Try-Catch blocks to handle common network issues, such as LDAP connection timeouts or "Access Denied" errors during sensitive attribute enumeration.
- **Documentation within Code**: Comprehensive comments were added to explain complex LDAP filters and the logic behind categorizing AD objects.

4.2 Version Control & Collaboration (GitHub)

The project is hosted and managed using Git to track changes and maintain version integrity:

- **Repository**: The project is publically hosted on GitHub, providing a centralized location for the source code and documentation.

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GhostPath is designed as a standalone tool for ease of deployment in security environments:

A. System Requirements

- **Operating System**: Windows 10/11 or Windows Server (2016, 2019, 2022).
- **Software Dependencies**: PowerShell 5.1 or higher (or compiled .NET runtime if using the executable version).
- **Permissions**: Must be executed in the context of a domain user with read access to Active Directory.

B. Installation & Configuration

1. Clone the repository from GitHub: git clone

https://github.com/AkmalMohamed7/GhostPath-tool

2. Navigate to the project directory.
3. No external installation is required as the tool uses native Windows protocols.

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The tool is executed via the command line with various parameters:

- **General Scan**: .\GhostPath.ps1 (Runs a full domain enumeration).
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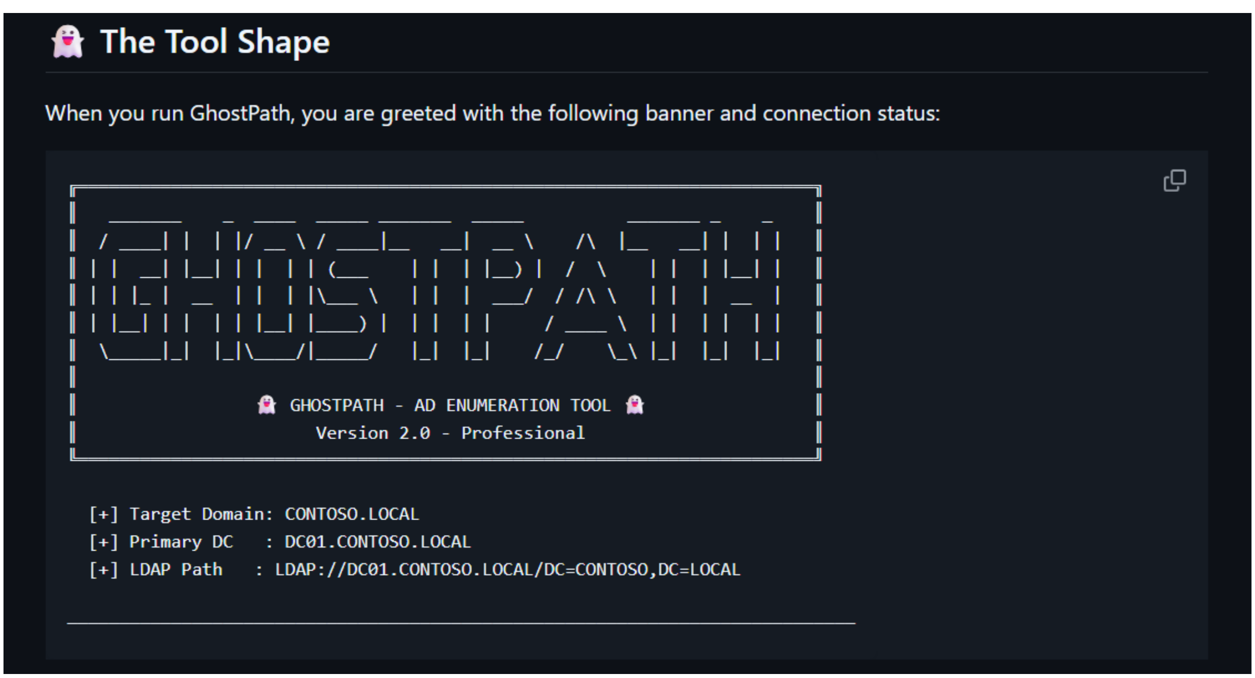

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5. Results & Tool Output

This section demonstrates the actual output of the GhostPath tool when executed in a controlled Active Directory lab environment.

The screenshots below highlight the tool's ability to enumerate domain objects and present them in a structured, high-contrast format for security analysis.

5.1 Full Domain Enumeration (Default Execution)

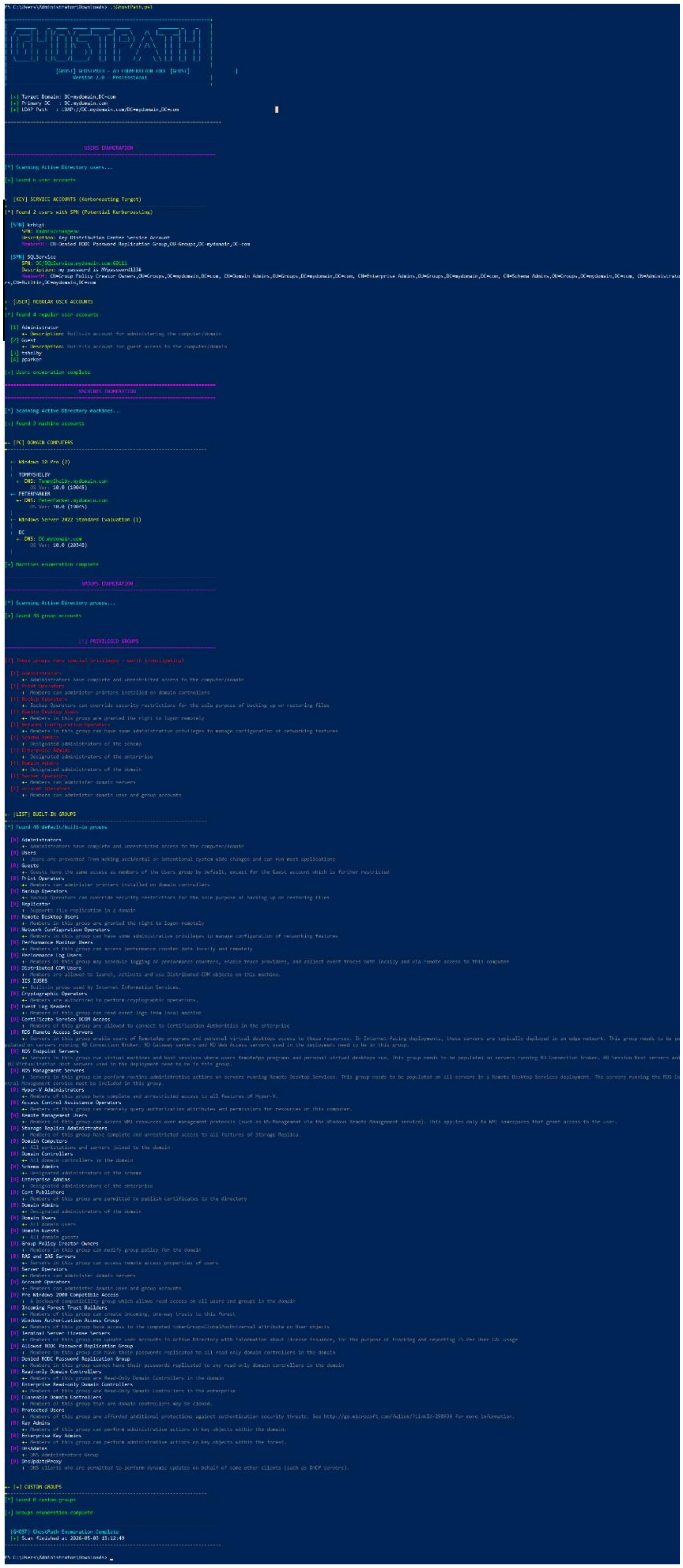
This screenshot shows the default execution of the GhostPath tool using the following command:

.\GhostPath.ps1

In this mode, the tool performs a complete enumeration of the Active Directory environment, including:

- Domain Users
- Domain Machines
- Security Groups
- Privileged Groups
- Kerberoastable Accounts

The output is presented in a structured, high-contrast format, allowing security professionals to quickly identify critical information such as domain controllers, high-privilege accounts, and potential attack targets.


PS C:\Users\Administrator\Downloads> .\GhostPath.ps1
Version 2.0 - Professional
[+] Target Domain: DC=mydomain,DC=com
[+] Primary DC : DC.mydomain.com
[+] LDAP Path : LDAP://DC.mydomain.com/DC=mydomain,DC=com
USERS ENUMERATION
[*] Scanning Active Directory users...
[KEY] SERVICE ACCOUNTS (Kerberoasting Target)
[*] Found 2 users with SPN (Potential Kerberoasting)
[SPN] krbtgt
Description: Key Distribution Center Service Account
MemberOf: CN=Denied RODC Password Replication Group,OU=Groups,DC=mydomain,DC=com
[SPN] SQLService
Description: my password is MYpassword1234
[USER] REGULAR USER ACCOUNTS
[*] Found 4 regular user accounts
[1] Administrator
[2] Guest
[3] tshelby
[4] pparker
MACHINES ENUMERATION
[*] Scanning Active Directory machines...
[PC] DOMAIN COMPUTERS
Windows 10 Pro (2)
TOMMYSHELBY
PETERPARKER
Windows Server 2022 Standard Evaluation (1)
DC
GROUPS ENUMERATION
[*] Scanning Active Directory groups...
PRIVILEGED GROUPS
[+] CUSTOM GROUPS
[GHOST] GhostPath Enumeration Complete
[+] Scan finished at 2026-05-03 15:12:49
PS C:\Users\Administrator\Downloads>

5.2 User Enumeration & Kerberoasting Detection

This screenshot shows the result of running:

.\GhostPath.ps1 -ObjType U

The tool successfully enumerates domain users and identifies service accounts with Service Principal Names (SPNs), which are potential targets for Kerberoasting attacks.

Special attention is given to:

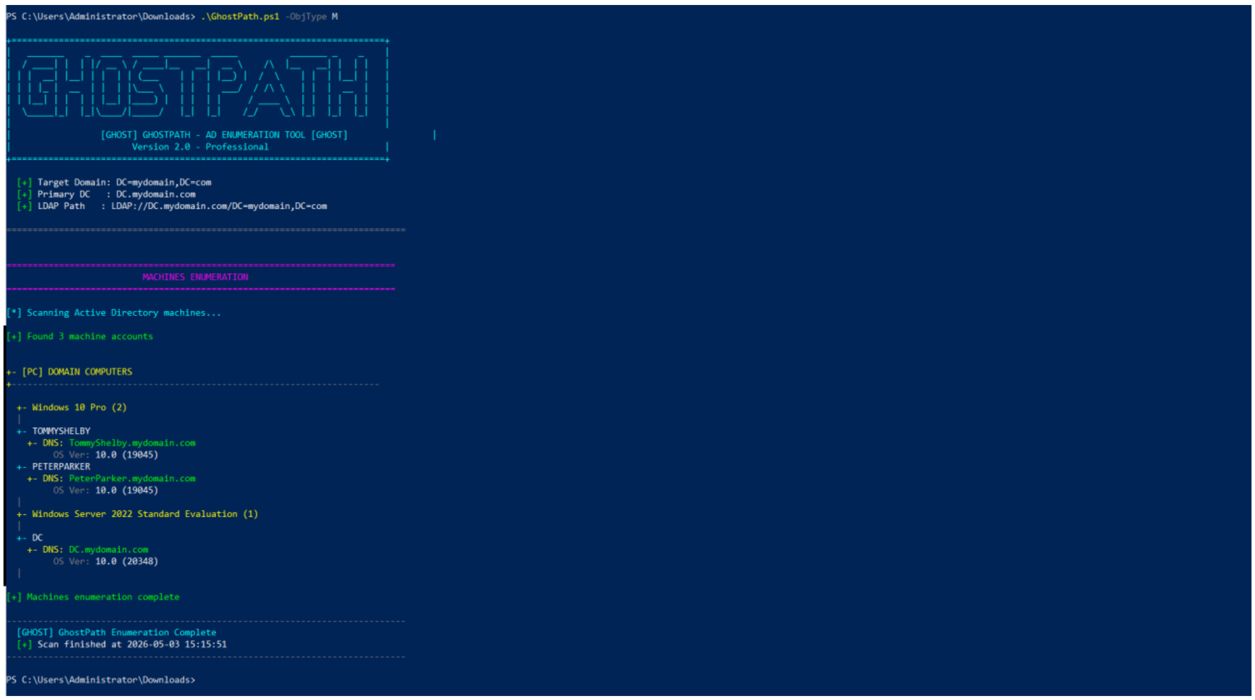
- Accounts with SPNs
- Descriptions and group memberships
- Highlighted security-relevant attributes

5.3 Machine Enumeration

This screenshot demonstrates the execution of:

.\GhostPath.ps1 -ObjType M

The tool categorizes domain machines based on their operating systems, allowing security analysts to quickly identify outdated or vulnerable systems within the environment.

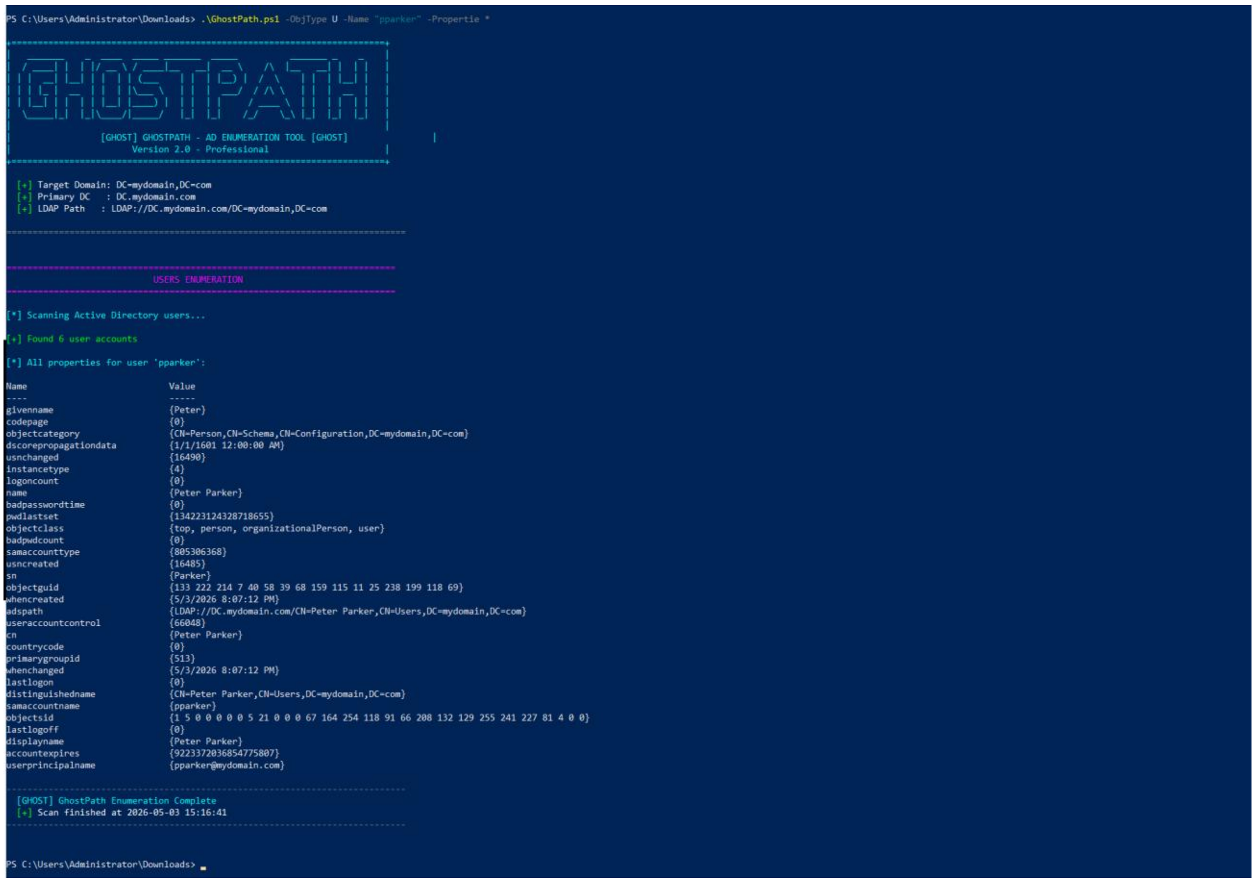


5.4 Detailed Object Inspection

This screenshot shows the execution of:

.\GhostPath.ps1 -ObjType U -Name "pparker" -Propertie *

In this mode, the tool retrieves and displays all available LDAP attributes for a specific object, enabling deep inspection and analysis of user properties.



6. Conclusion

This project presented the design and implementation of **GhostPath**, an Active Directory enumeration tool tailored for security professionals and red teamers.

The tool successfully achieves its primary objectives by:

- Automating the enumeration of domain users, machines, and groups
- Identifying security-critical elements such as privileged groups and Kerberoastable accounts
- Providing a structured and high-contrast command-line interface for efficient analysis

Through practical testing in a controlled lab environment, GhostPath demonstrated high accuracy, stability, and performance when interacting with Active Directory over LDAP.

Overall, the project highlights the importance of automation in security assessments and showcases how well-structured tooling can significantly improve visibility into domain environments.